

MUSTAFA AMIR

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Education

Asia Pacific University of Technology and Innovation (APU)

Nov 2022 – Jun 2022

Bachelor of Science in Information Technology

Kuala Lumpur

- **Courses Taken:** Introduction to Programming (Python & C), Object Oriented Programming (Java), Web Development (HTML, CSS, JS, ASP.NET & C#), Programming for Data Analytics (R), Database Management (MS SQL Server), Cloud Computing (AWS S3, ECS, EC2, VPC & API Gateway)

Summary of Skills

Languages: Python, Java, C, C#, JavaScript & TypeScript, HTML & CSS, .NET, SQL, R, Dart, GraphQL
Frameworks: FastAPI, Flask, Spring & Spring Boot, ASP.NET, React (Router & Next.js), Vue, Flutter, Node.js
Tools: Docker, Kubernetes, Git & Github, Jenkins, Maven, PostgreSQL
AI/ML: PyTorch, Tensorflow, scikit-learn, Keras
Platforms: Linux (Ubuntu), WSL, AWS, GCP, Supabase, Vercel
Soft Skills: Technical Leadership, Ownership & Accountability, Problem Solving, Adaptability

Experience

R&D Application Development Intern

May 2025 – Sep 2025

Continental AG

Petaling Jaya and Hanover (remote)

- Designed and shipped a task management dashboard in React & TypeScript, building custom drag-and-drop logic, optimized for complex state transitions and large datasets
- Re-architected component hierarchy to eliminate prop drilling, implementing centralized state via Context & reducer patterns to support scalable feature expansion
- Integrated with Spring Boot microservices through REST APIs, implementing optimistic updates, rollback handling, and local persistence to ensure resilience under intermittent network conditions
- Operated within an existing CI/CD pipeline (GitHub Enterprise & Jenkins), writing unit tests and enforcing code review standards to maintain production reliability

Technical Co-founder

Nov 2022 – Dec 2023

Katalog

Kuala Lumpur

- Led product strategy and development of a social platform connecting users with F&B establishments, looking at RM150,000 in potential funding pre exit
- Onboarded 54 restaurant partners for initial trials, grew user base to 500 active trial users, developed and tested subscription-based revenue model (MRR)

Highlighted Projects

OpenPulse

[Link](#)

- In my spare-time, I am building OpenPulse — an open-source, multi-tenant event intelligence platform.
- Stack: React, Zustand, FastAPI, FastAPI WebSockets, SQLAlchemy & Alembic, PostgreSQL, Redis, Docker

YOLOv8 Brain Tumor Segmentation & Classification Pipeline with FastAPI for Model Serving

[Link](#)

- Built an end-to-end medical imaging pipeline that trains YOLOv8 on the BRISC 2025 brain tumor MRI dataset for simultaneous classification and instance segmentation, as part of my **final year project**
- Stack: YOLOv8, FastAPI, OpenCV, Pillow, PyTorch, Tensorflow, pandas

Drug-Allergy Clinical Decision Support (CDS) Microservice

[Link](#)

- Built a Node.js/Express microservice that intercepts FHIR MedicationRequest webhook events and blocks prescriptions that conflict with a patient's active allergies
- Stack: Node.js 20, TypeScript, Express 5, Zod, Axios, Vitest, HAPI FHIR, RxNav.

Publications

- Mustafa Amir, Kohila Malar Kalesamy. "YOLOv8-Based Instance Segmentation for Accurate Organ and Lesion Delineation in Radiological Images." *International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2026.